Module 1: Fundamentals of Differential Equations

Welcome to **Module 1**. Before diving into complex solving techniques, it is essential to understand the basic terminology, classifications, and origins of differential equations. This module lays the groundwork for the entire course.

Module Learning Objectives

By the end of this module, you will be able to: 1. Define a differential equation and distinguish between Ordinary (ODEs) and Partial (PDEs) Differential Equations. 2. Accurately identify the **Order** and **Degree** of any given differential equation. 3. Construct a differential equation from a general mathematical function by systematically eliminating arbitrary constants.

Topic Index

Navigate through the topics covered in this module using the links below:

1. Basic Definitions

- **What is a Differential Equation?**
- **Ordinary Differential Equations (ODEs)** vs. **Partial Differential Equations (PDEs)**.
- Understanding dependent and independent variables.

2. Order and Degree of a DE

- **Order:** Identifying the highest differential coefficient.
- **Degree:** Determining the highest power of the highest derivative.
- **Rules:** Making the equation free from radicals and fractions before determining the degree.
- *Includes step-by-step solved exercises.*

3. Formation of Differential Equations

- The relationship between the number of arbitrary constants and the order of the resulting DE.
- **Methodology:** Differentiating ordinary equations and eliminating constants.
- *Includes step-by-step solved exercises.*

Next Steps: Start with the first topic, Basic Definitions, to begin your study of differential equations.